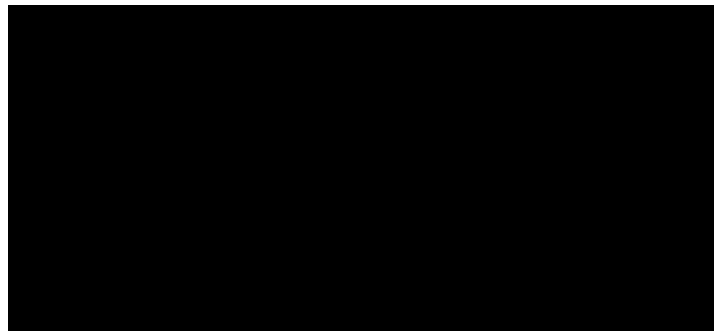


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Geo-Doodle

Group 2017.30



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1. High Level Description

1.1 Motivation

The world's urban population has been growing with increasing world population. In 1950, about 700 million resided in the urban areas, which is approximately 30% of the world population of 2.5 billion at the time [1]. The ratio of the urban population has now increased to 54% with the world population of 7 billion. By 2050, the proportion of the urban population is expected to be 66% with the projected world population of 9.6 billion [2]. In parallel with the increasing urban population, the role of urban planning continues to stand out [3].

During the urban planning, a scale model is often used to present and demonstrate the strategies and concept involved in the design; models effectively simulate the reality with great details which can be used as tools in the design process or as presentation tools [4]. Despite the benefits, models associate lead time for fabrication and cost, for each additional landscape. Besides, the models are not reusable for different projects that require different landscapes. They are static, and cannot utilize simulated effects that can demonstrate dynamic conditions of the landscape. The project that this report entails intends to resolve such problems that the existing solutions possess.

1.2 Project Objective

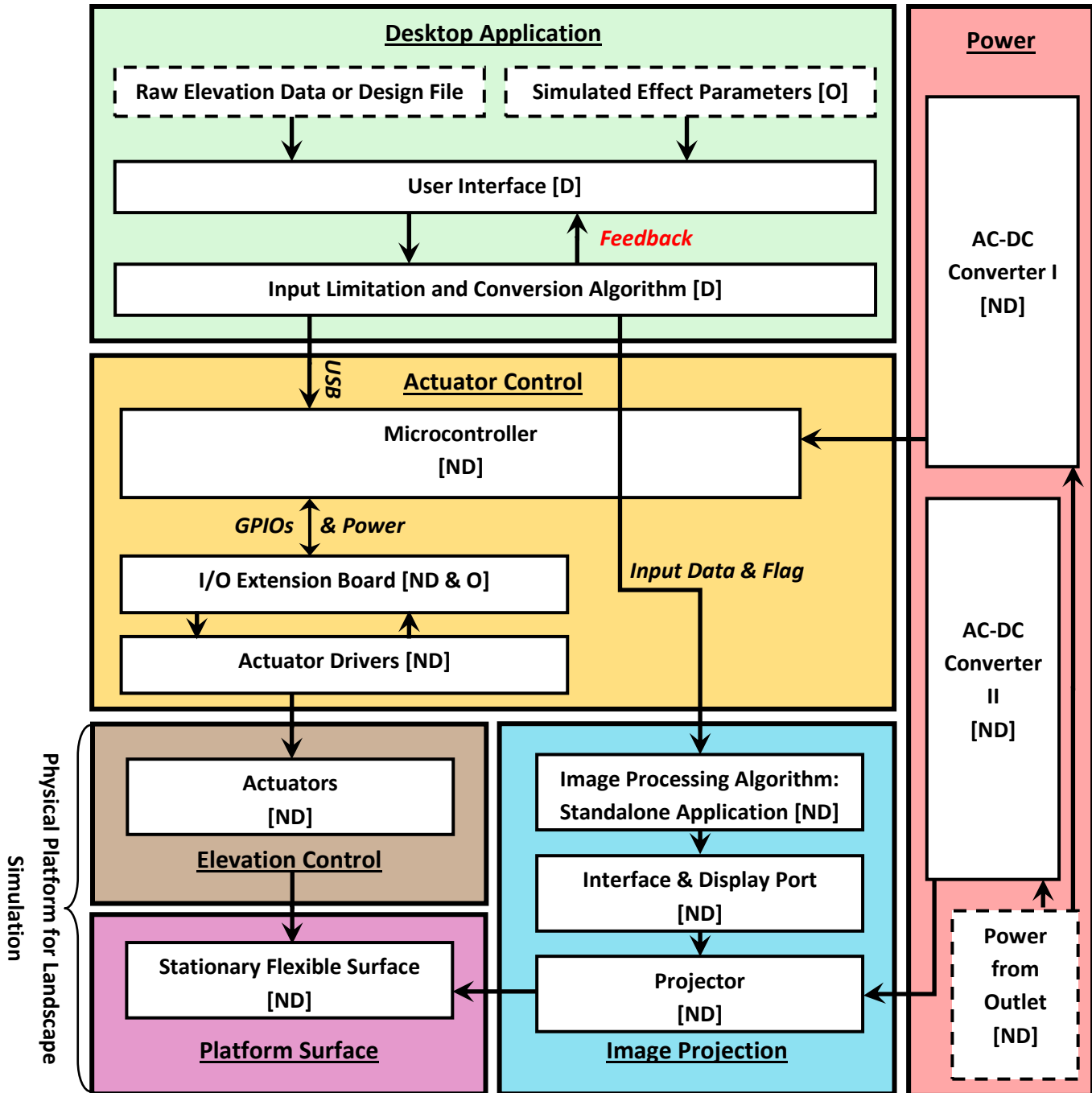
The objective of the project is to design a physical platform with a controllable surface that can change its shape promptly to help visualizing various landscapes during urban planning processes.

1.3 Block Diagram

The proposed solution uses a set of elevation data to configure elastic surface using limited number of actuators, which are either allocated in an n -by- n matrix formation. The simulated effects utilize a projector and require image processing techniques to minimize distortion perceived by the observers. The graphical representation of the system is shown in Figure 1. The different components from the block diagram in Figure 1 are presented in Table 1 below.

Table 1. Description of block diagram components

Component	Designed?	Description
Raw Elevation Data or Design File	Yes (for demonstration purpose)	Either of a set of raw elevation data or a design file is used as an input for landscape configuration.
Simulated Effect Parameters	Yes (for demonstration purpose) & Optional	Used as a set of input parameters to the image processing algorithm (e.g., precipitation rate and drainage rate of the system in the urban area)
User Interface	Yes	Front-end desktop application that collects inputs and deliver them to appropriate components.
Input Limitation and Conversion Algorithm	Yes	Elevation data is gated by the algorithm, which ensures that inputs are within the hardware limitation (i.e., number of actuators, flexibility of the surface). Elevation data is also converted into a set of information that can be directly applied to controlling actuators (as in voltage or running time of actuators).
Microcontroller	No	Controls the behaviour of the actuators. Uses a readily available product in the market.
I/O Extension Board	No & Optional	Provides more I/O ports, needed to drive multiple motors. Uses an available product.
Actuator Drivers	No	Provides sufficient current or force to drive actuators. Uses a readily available product in the market.
Actuators	No	Controls the elevation of the surface. Investigated further in the design section
Stationary Flexible Surface	No	Stretchable surface that simulates the landscape of interest. Uses a readily available product.
Image Processing Algorithm: Standalone Application	Yes	Uses the inputs to minimize distortion perceived by observers. Digitized information is sent to the projector.
Interface & Display Port	No	Allows communication from the algorithm to the projector using the standard medium (e.g., RS-232, HDMI, etc.). Uses an available product.
Projector	No	Projects images over the surface. Uses an available product.
Power from Outlet	No	The original source of power.
AC-DC Converter I	No	Transforms the AC power into DC power; power source of the projector.
AC-DC Converter II	No	Transforms the AC voltage into chip-level DC voltage; power source for microcontroller and drivers.



Legend

D: Designed by the team

ND: Not designed by the team; an existing solution will be implemented

[] : Input

O: Optional; may or may not be required and may or may not be implemented

Figure 1. High-level block diagram of Geo-Doodle

2. Project Specification

2.1 Functional Specifications

The functional specifications of Geo-Doodle are shown in Table 1. The specifications are classified as essential and non-essential based on their importance and significance within the system.

Table 2. Functional specifications of Geo-Doodle

1. Surface Height	Essential
The surface used for landscape simulation must be able to extend by maximum +10 cm in vertical direction. The uncertainty of the stretched surface must be within the range of ± 1 cm with respect to the desired vertical displacement. The surface must support the slope range of 0 to 1.	
2. Landscape Configuration Time	Essential
The simulated landscape must be completely configured within 3 minutes from the initiation.	
3. Projected Image Resolution	Essential
The effective resolution (perceived by the observers) of the projected image (emitted from a projector) must be at least 512 pixels in width 512 pixels in length, so the image is capable of representing sufficient granularity.	
4. Synchronization of Surface and Projected Image	Essential
Projected image must respond to the changing landscape within 100 milliseconds [4]; the maximum latency of the projected image with respect to the landscape must be 100 milliseconds.	
5. Input Limitation for Landscape Simulation	Essential
The system must monitor the landscape datasets and evaluate whether they can be represented by the hardware (motors and surface). The inputs must be processed only if they are found to be valid (within the range of surface height, resolution of 512x512). The evaluation result is notified to users.	
6. CAD Compatibility	Non-Essential
To simulate landscape, the raw elevation data can be replaced by a computer-aided design file, which contains comprehensive information of a geometrical shape designed, including elevation.	
7. Distortion of Projected Image	Essential
The image must not be distorted with the elevated landscape, simulated by surface.	

2.2 Non-functional Specifications

The non-functional specifications of Geo-Doodle are presented in Table 2. Similar to how functional specifications were classified, the non-functional specifications are classified as essential and non-essential based on their importance and significance.

Table 3. Non-functional specifications of Geo-Doodle

1. Surface Colour	Essential
The surface colour must be light (white, bright yellow) for maximal reflection of projected images.	
2. Platform/Surface Size	Essential
The dimension of the platform and the surface is 60 cm in width by 60 cm in length.	
3. Elevation Actuator Dimension	Essential
Elevation actuators are located in a square matrix format of 5 by 5 on the platform. Maximum size of elevation actuator is 10cm in width, 10cm in length or 10cm in diameter depending on actuator shapes.	
4. Projector Location	Essential
The vertical location of the projector must be at least 30 cm above the maximum height achieved by the surface, to achieve the specified resolution and so the surface can be covered completely with projected images.	
5. Cost of the system	Non-Essential
The cost associated with the system does not exceed \$750.	
6. Weight of the system	Non-Essential
The weight of the system does not exceed 20 kg.	
7. Surface Elasticity	Essential
The surface material must be hyperelastic or materials that are nearly elastic as hyperelastic material, so it can recover its original state when force exerted onto it is removed.	

3. Detailed Design

3.1 Surface Elevation Control

The inner platform of Geo-Doodle is where elevation actuators integrated on a wooden platform operate to simulate 3D visualization of an image. An image data from UI desktop application through microcontroller along with control algorithm is fed to the actuators to simulate the elevation accordingly. The primary focus and specifications for the inner platform subsystem design are summarized as below.

Figure 2 below shows top and side views of the elevation actuators integrated wooden platform with measurement specifications indicated.

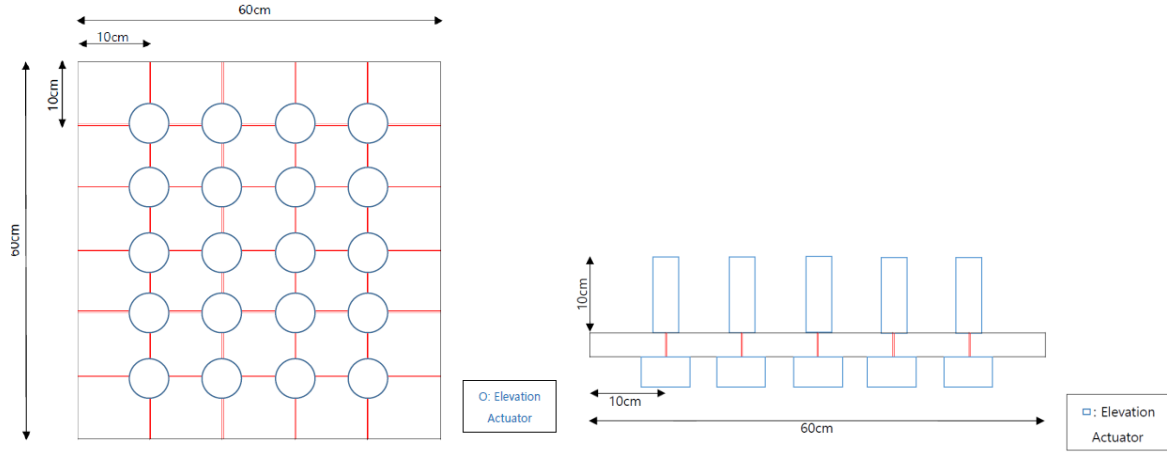


Figure 2. Platform Top, Side Views

3.1.1 Surface Elevation Control

There are two platform elevation control design alternatives that are considered: synchronous elevation control and asynchronous elevation control. Synchronous elevation control alternative for the platform design is to control all the elevation actuators simultaneously. This is achieved by placing an actuator controller (ex. Motor) to the elevation actuator for all individual actuators. Asynchronous elevation control alternative for the platform design is to control the elevation actuators individually. This is achieved by adopting customized adaptive robots to move around the inner platform with implemented algorithm to selectively control elevation actuators.

Selection of the elevation control alternative is performed through decision matrix with evaluation criteria formulated based on the specifications. The evaluation criteria along with weightings are shown in Table 4 below and decision matrix is computed in Table 5.

Table 4. Evaluation Criteria for Elevation Control Alternatives

Evaluation Criteria	Weightings (%)
Design Complexity	35
Cost	25
Max Landscape Configuration Time	25
Max Power Consumption	15

Design complexity of asynchronous elevation control alternative is higher since it requires 'Navigation & Path-Finding Algorithm' implementation for the adaptive robots in order to control their paths effectively. In addition, accuracy of placing the robot right next to the elevation actuator is important which may be challenging. On the other hand, synchronous elevation control alternative does not require algorithm implementation which reduces design complexity. Relative complexity is recorded in the decision matrix to be 3 and 8 for synchronous elevation control and asynchronous elevation control alternatives respectively.

The number of actuator controller required for synchronous elevation controlled platform is equal to the number of elevation actuators while asynchronous elevation controlled platform can use

minimum a single adaptive robot instead. A single adaptive robot is more expensive than a single actuator controller which may be motor or hydraulic cylinder. Though, few number of adaptive robots are in general cheaper compare to 25 actuator controllers. For decision matrix computation, each actuator controller for synchronous elevation control is determined to be \$11 [5] which totals up to \$275 for the platform and a single adaptive robot price is determined to be \$100 [6]. The costs used for the computation are found from electronic stores online with components that are suitable for the platform design.

Asynchronous elevation control consumes higher landscape configuration time compare to synchronous elevation control. For instance, a single adaptive robot used for asynchronous elevation control to elevate all the actuators to its full height takes 25 times that of the synchronous elevation control mechanism takes to do the same job. Landscape configuration time can be decreased by utilizing multiple adaptive robots but the minimum landscape configuration time converges to the synchronous case with significant cost increase. For decision matrix computation, time consumed to elevate all elevation actuators to their full heights are computed with an assumption that it each actuator controller and an adaptive robot takes 5 seconds to elevate an elevation actuator to the maximum height.

Synchronous elevation control consumes larger power compare to asynchronous elevation control. For instance, with an assumption that power consumed to control each actuator controller is equal to the power consumed to control an adaptive robot, power consumption for synchronous elevation control is approximately 25 times that of the asynchronous case. For decision matrix computation, the data used to control a single actuator controller and an adaptive robot is 4.5W [5] and 9W respectively. For synchronous elevation control platform, the total power consumption sums up to 112.5W.

It is to be noted that decision matrix computation is performed to result that lowest total scored alternative is the most preferable choice.

Table 5. Decision Matrix Computation for Platform Elevation Control Design Alternatives

	Synchronous Elevation Control			Asynchronous Elevation Control		
	C _{i1}	r _{i1}	S _{i1}	C _{i2}	r _{i2}	S _{i2}
Design Complexity (35%)	3	0.375	13.125	8	1	35
Cost (25%)	275	1	25	100	0.3636	9.091
Max Landscape Configuration Time (25%)	5	0.04	1	125	1	25
Max Power Consumption (15%)	112.5	1	15	9	0.08	1.2
Totals	54.125			70.291		

From decision matrix computation, synchronous elevation control is a preferable alternative. In the next subsections, elevation actuator design alternatives that are suitable to synchronous elevation controlled platform are discussed.

3.1.2 Hydraulic Cylinders – First Iteration

The first alternative considered for elevation actuator is hydraulic cylinder. Hydraulic cylinders used as elevation actuators are visualized as shown in Figure 3 below.

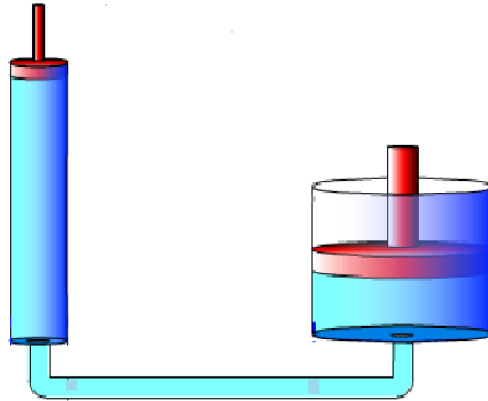


Figure 3. Elevation Actuator Design Alternative: Hydraulic Cylinder

Hydraulic cylinders piston is pushed upward as downward pushing force is applied at the reservoir piston that pushes the water inside the tank to fill the hydraulic cylinder. Oppositely, upward pulling force is applied at the reservoir piston to lower the cylinders piston's height. Such hydraulic system is a viable solution but drawbacks exist. The hydraulic system used for the platform should be limited to electric-driven (voltage inputs) component so that it can be operated from electric power supply. If non electric-driven hydraulic system is to be used, custom design is required at the inputs of hydraulics to achieve electric-driven mechanism which may increase complexity of the design. The availability of electric-driven hydraulic system that satisfies the platform dimension spec is low and the cost and weight is high compare to other feasible alternatives such as stepper motors. In addition, due to incompetency of the group members in mechanical design especially in hydraulic systems, fabrication of hydraulic system integrated platform may be challenging.

3.1.3 Stepper Motor with Screw – Second Iteration

The second alternative considered for elevation actuator is stepper motor with screw as shown in Figure 4 below.

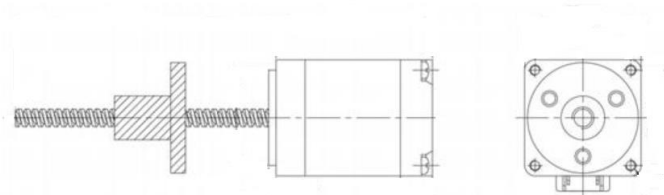


Figure 4: Elevation Actuator Design Alternative: Stepper Motor with Screw [7]

Stepper motor with screw is a motor with built-in screw connected to its shaft. Stepper motor with screw spins through screw hole to control the elevation of the platform. Since group members are

more competent and experienced utilizing motors, the design implementation is more familiar than hydraulic system implementation on the platform. Also, cost and weight of stepper motor with screw is low and availability is high that it could be easily found in electronic stores. However, there is a slight increase in design complexity since the platform holes must be designed in screw shape. Also, the platform has low durability because wooden platform may wear out as the screw is screwed back and forth through the hole.

3.1.4 Gear Controlled Poles – Third Iteration

The third alternative considered for elevation actuator for the platform is gear controlled poles as shown in Figure 5 below.

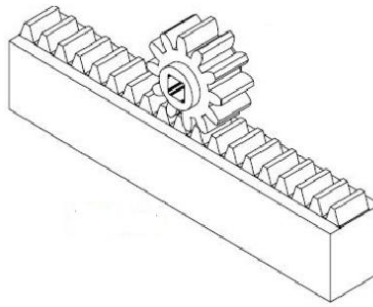


Figure 5. Elevation Actuator Design Alternative: Gear Controlled Poles [8]

Gear controlled pole is a combination of gear and a sawtooth-shaped pole interlocked with the gear. The gear connected to a motor rotates the gear to control the elevation of the pole. Stepper motor is a suitable component to be connected to the gear. The design complexity of gear controlled poles on the platform is low since appropriate motor connection to the gear is not a challenging implementation. The durability of the elevation actuator is moderate since operation of the gear on the sawtooth-shaped pole is likely to damage gear and the pole to make it dull. Cost and weight of this elevation actuator is low and availability is high since the gear components and the shaped-poles are commonly found.

3.1.5 Elevation Actuators Design Alternative Selection

Design alternatives for elevation actuators that are discussed in previous subsections are evaluated on subjective criteria along with weightings listed below in Table 6 by revisiting the decision matrix technique to choose the best alternative as shown in Table 7.

Table 6. Evaluation Criteria for Elevation Actuators

Evaluation Criteria	Weightings (%)
Design Complexity	30
Durability	20
Cost	15
Weight	10
Availability	10

It is to be noted that decision matrix computation is performed to result that lowest total scored alternative is the most preferable choice. The design complexity, durability and availability criteria for an alternative are computed with relative scores to other alternatives. For cost and weight, appropriate model of each alternative is found from online electronic store and used for comparison. Hydraulic cylinder cost found suitable for the platform is \$60 [9] each which totals up to \$1500 for 25 elevation actuators. Stepper motor with screw and gear controlled poles are \$15 [10] and \$30 [11] each. Weight of a small hydraulic cylinder, stepper motor with screw and gear controlled pole used for the platform is near 500g, 200g and 300g respectively.

Table 7. Decision Matrix Computation for Elevation Actuator Design Alternatives

	Hydraulic Cylinder			Stepper Motor with Screw			Gear Controlled Poles		
	C _{i1}	r _{i1}	S _{i1}	C _{i2}	r _{i2}	S _{i2}	C _{i3}	r _{i3}	S _{i3}
Design Complexity (35%)	8	1	35	2	0.25	8.75	4	0.5	17.5
Durability (25%)	2	0.333	8.333	6	1	25	5.5	0.9167	22.917
Cost (15%)	1500	1	15	375	0.25	3.75	750	0.5	7.5
Weight (10%)	12500	1	10	5000	0.4	4	7500	0.6	6
Availability (10%)	7	1	10	2	0.286	2.857	2	0.286	2.857
Totals	78.333			44.357			56.774		

From the decision matrix, stepper motor with screw is determined to be the most preferable design alternative for elevation actuators.

3.1.6 Inner-Platform Design

From the above subsections, elevation control mechanism and elevation actuator type are determined through quantitative analysis by the use of decision matrix computation. In this part, detailed design plan for the inner-platform implementation is discussed with the chosen design alternatives.

For elevation control, bipolar stepper motor with screw is integrated on the platform following the format described in Figure 2. Total of 25 stepper motors with screws are required for the platform as elevation actuators. As raw elevation data is provided as an input to the UI desktop application, microcontroller subsystem sets the motor operation time to the Arduino inputs to control the stepper motors screw elevations.

For appropriate stepper motor selection, elevation actuator's dimensions, weight and rotation torque are considered based on the specifications. Other than aforementioned specifications, rotational torque of the stepper motor should be considered so that the motor provides high enough torque to overcome opposing frictional torque. In order to calculate the frictional torque against the rotational torque applied to the actuator, free body diagram of an actuator on the platform is represented as below in Figure 6.

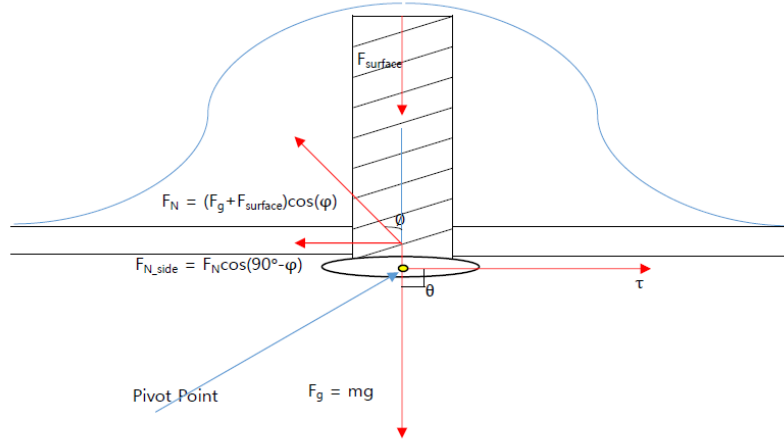


Figure 6. Free-Body Diagram of Stepper Motor with Screw on the Platform

From the FBD in Figure 6, the frictional torque is calculated as follows. Using the general torque equation shown in equation (1), the equation is developed to the screw situation as shown in equation (2) where r_s is the radius of the screw, F_{N_side} is the normal force on the frictional surface and μ is friction coefficient of two materials in contact which in this case is steel-wood.

$$\tau = rF\sin(\theta) \quad (1)$$

$$\tau_{friction} = r_s \sin(\theta) (F_{N_side} \mu) \quad (2)$$

Since the angle between the pivot point and the applied torque is 90 degrees, $\sin(\theta) = 1$ for equation (2). Also, it is to be noted that normal force is contributed by the gravitational force and opposing reaction force from the surface leading to equation (3) as below.

$$\tau_{friction} = r_s F_N \cos(90^\circ - \phi) \mu \quad (3)$$

In order to find a stepper motor with screw that meets this specification, iterative analysis should be performed for several models of stepper motor with screw that are found and the final model chosen has the specification as summarized in Table 8 below.

Table 8. Stepper Motor with Screw Final Model Specification

Screw Radius: r_s (cm)	0.3
Screw Length: L_s (cm)	10
Dimensions (W x L) (cm)	4.23 x 3.8
Weight (kg)	0.200

The model satisfies the elevation actuator dimension spec limited to 10cm in width and length, screw length minimum of 10cm and weight less than 20kg which is 5kg in total.

Frictional torque is calculated using the equations (1) to (3) derived with the motor model specification provided in Table 8. This frictional torque was compare to motor torque to validate that the frictional torque is less than the motor torque. It is to be noted that the mass of the stepper

motor is used by dividing the weight by gravity acceleration of 9.8m/s^2 . Also, the friction coefficient between steel-wood contact is found to be 0.5 [12]. Lastly, the screw tooth was assumed to have angle of 30° .

$$\tau_{friction} = 0.003((0.0204 \times 9.8) + 6) \sin(30^\circ) 0.5 = 0.00465 \text{ Nm}$$

Therefore, the chosen actuator model must have rotational torque greater than 0.00465Nm .

The last part to be discussed for inner-platform subsystem design is the attachment of the elevation actuators to the surface. It is found in the design planning process that direct attachment of the screw to the surface cause problems since the screw rotates while being elevated also makes the surface to rotate in the same direction as the screw. In order to prevent this, separate cap that connects the surface to the screw was designed. Therefore, the rotating screw is not affecting the surface.

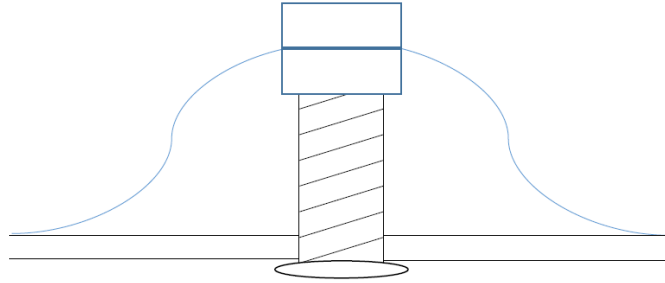


Figure 7. Surface to Screw Attachment Design by Use of a Cap

3.2 Platform Surface

3.2.1 Surface Material Selection

Selecting surface material of the project's platform is one of the most important decisions that need to be made. Ability to be stretched is a crucial feature that the material needs to have since stepper motors with screw change their height, the surface material should be stretched. Density or weight of the material is an important factor since it directly affects the required force from stepper motors with screw. Lastly, availability of highly reflective colour is another important factor since the image is projected onto the surface material. The team finds that spandex, latex, knit and nylon are possible options as surface material since they meet the basic factors. Below table shows the basic specifications for the four materials.

Table 9. Specification of Spandex, Latex, Knit and Nylon [13][14][15][16]

Material	Elasticity	Density	Availability of Highly Reflective Colour	Price
Spandex	500 %	1.15 to 1.32g/cc	Yes	\$20 / yard
Latex	300 %	0.06g/cc	Yes	\$13 / yard
Knit	150 %	1.31g/cc	Yes	\$16 / yard
Nylon	120 %	1.15g/cc	Yes	\$34 / yard

To figure out which material is the best for the project, decision matrix is made. Below table is the decision matrix for choosing the best material.

Table 10. Decision Matrix of Spandex, Latex, Knit and Nylon

Material	Elasticity	Availability of Highly Reflective Colour	Density	Price	Score
Weights	40%	30%	20%	10%	
Spandex	5	5	4	4	4.7
Latex	3	5	5	5	4.2
Knit	1	5	4	5	3.2
Nylon	1	5	4	3	3

Since the major criteria of surface material for the project is elasticity, 40% of the weight is given to elasticity. Availability of highly reflective colour of the surface material is chosen as a factor to be considered since it needs to reflect projected image instead of absorbing it, therefore 30% of weight is given. Density of the material is also compared, since a material with higher density may exert more gravitational force to controlling plant beneath the layer. Finally, price is also an important criterion in this project, since it is a factor which must be optimized as much as possible in a good engineering design.

Spandex has the highest elasticity among the materials under comparison of 500%, which means that it can stretch up to 500% of its original length. Therefore, it offers the highest range of representation of altitude. Latex has the second highest elasticity of 300%, which is reasonably good but not the best option due to existence of spandex. Knit and nylon are also under consideration of being a potential material for this project, but they offer less than the elasticity of 300% that latex offers. Therefore, score of 5 is awarded to spandex, 3 to latex and 1 to knit and nylon.

Colour of the surface determines visibility of the projected image on the surface. Basic rule of optics say that white surfaces reflects incoming lights, while darker surfaces absorb lights. Therefore, to maximize visibility of projected image, it is crucial to use the whitest surface available in the market. All of the materials under this comparison can be purchased in light colour, and therefore score of 5 was awarded for every material.

In density category, latex is awarded the highest score with its density of 0.06 g per cubic centimeter. Therefore, it applies the least amount of stress to the control system, which allows precise control of the altitude representing poles, and potentially reduces power consumption of the servo motors. The other materials have densities ranging from 1.15 g/cc to 1.32 g/cc, which is 20 times higher than that of latex. However, the absolute value of their densities is still significantly lower than the point where the surface may apply enough stress to the control unit to cause malfunction. Therefore, latex is awarded a score of 5, while the rest are awarded a score of 4.

Price is always a key factor in engineering design and must be compared. For consistency, the prices of the materials are taken from amazon. After researching, it is concluded that latex is the cheapest solution, which costs \$13 per yard. Nylon, on the other hand, is the most expensive

solution, which costs \$34 per yard. Knit costs \$16 per yard, and spandex costs \$20 per yards. Score of 5 was awarded for materials that cost below \$19, 4 for below \$29, and 3 for below \$39.

The above decision matrix shows that spandex has the highest total score. It is because it is awarded the highest score for elasticity, which has the highest weight among the factors, and also awarded good scores for other factors as well. Latex beats spandex in density and price by 1 point, but it is not good enough to beat spandex's elasticity. Knit and nylon are quite similar in terms of elasticity, density, and colour availability, while knit being cheaper and having a better score than nylon by 0.2 points. In conclusion, spandex is awarded the total score of 4.7, while latex, knit, and nylon winning 4.2, 3.2, and 3 points respectively.

3.2.2 Spandex Fabric Selection

There are three different spandex fabrics that meet surface specifications that the team finds in local store. Not all spandex fabrics meet the specification in terms of highly reflective colour. Team finds three fabrics that are light enough in terms of colour so that the projected image is shown clearly. The team measures elasticity of all three feasible spandex fabrics and finds that they are all different. For the purpose of the project, since the stepper motors are the ones that pushes the fabric surface to the desired height, fabrics that need less force when they get stretched are considered to be better spandex fabrics. The team measures the force that stepper motors with screw need to stretch one point of fabric up to 10cm, which is maximum height the stepper motor with screw goes up, and to maximum height. The maximum stretchable heights are different for all three fabrics. However, for the purpose of this project, as long as the maximum stretchable height of measuring fabric is more than 10 cm, it is acceptable. Force is measured with force metres and 20 cm x 20 cm spandex fabrics are chosen as samples. The dimension of 20 cm x 20 cm is chosen as a sample because the distance between the stepper motors with screw is 10 cm apart and the team is measuring force of the center of the fabric. The middle points of four sides which represent neighbor stepper motors are to be held firmly. Below table shows the comparison of the three spandex fabrics.

Table 11. Comparison Table of 3 Different Spandex

	Spandex 1	Spandex 2	Spandex 3
Maximum stretchable height	22.5 cm	24.3 cm	22.7 cm
Force required to stretch up to 10 cm	6.9 N	6.5 N	6.2 N
Force required to stretch up to maximum height	28.8 N	29.5 N	25.1 N

Each forces and maximum stretchable height are measured 5 times each and the table shows the average of the five measurements. From the table the team decides that Spandex 3 is the best choice for the projects. Although maximum stretchable height for spandex 3 is the lowest of the three, force required to stretch it up to 10 cm is also lowest. Therefore, spandex 3 is chosen as the best surface material for the project.

3.3 Image Projection

3.3.1 Projector

The image projection requires a projector. The criteria of brightness, size, weight, cost, native resolution, and projection distance affect the selection of the projector. Due to the allocation of majority of the project budget to the platform and surface, only the projectors available in ECE inventory are considered for selection. Having more than one projector increases the complexity and cost of the project. Hence, the whole project is designed to meet the specifications with one projector. The projected image resolution must be at least 512 pixels by 512 pixels as specified in the specifications. Native resolution of the projector is considered as an essential criterion with weight of 30%. The score of this criterion will be 1 if the native resolution meets the specification, and it would be 0 if it does not meet the specifications. As the Geo-Doodle is supposed to use in presence of ambient light, the brightness of the projector is the one of the other essential criteria which has the weight of 30%. Projection distance criterion gets weights of 20% as the platform is supposed to use indoor, and projector cannot be mounted very far from the surface. Brightness less than 2000 lumens is only suitable for projection in darkened room so that the projected image is not washed out by ambient light. Projectors with this range of brightness get 0 score point. Projectors with brightness of 2000 lumens or higher get 1 score point as they are suitable for projection in presence of some room lights. The size and weight criteria each has 5% scoring weight as they do not affect the specifications. The weight of the projector is preferred to be low so that the mounting of it become easier. The volume of the projector is preferred to be small. This eases the mount of the projector on top of the surface. Low weight results in higher scores for this criterion. The acceptable dimensions are the ones result in a size less than the size of the surface ((wide x length) projector < size of surface). This criterion is scored by reciprocal of the volume. The cost has the weight of 10% and the projectors are scored with reciprocal of their costs. The purchased cost of the projectors is taken into consideration for this comparison. However, these costs are not included in the final cost calculation of the project as the selected projector is rented from the ECE inventory. The following table summarizes the characteristics of the candidate projectors as well as their scoring points. It can be seen that the BenQ projector [17] is chosen for the project with a higher score of 85% over 3M Mpro projector [18].

Table 12. Projector Selection

Projector	Native Resolution (30%)		Brightness (30%)		Projection Distance (20%)		Size (5%)		Weight (5%)		Cost (10%)		Total Score (100%)
	pi x pi	score	lumens	score	cm	score	cm ³	score	kg	score	CAD	score	
BenQ MP620c	1024 x 768	1	2000	1	93.35	1	9 x 28 x 22	0.03	2.7	0.06	916	0.47	85.18%
3M Mpro150	1280 x 800	1	15	0	106.68	0.875	13 x 6.1 x 2.3	1	0.16	1	432	1	67.50%

3.3.2 Image Processing

This section provides the requirements to meet the specifications while designing the algorithm and the algorithm techniques used for design. The image processing is done through language of MATLAB.

3.3.3 Requirements

3.3.3.1 Star Shape Surface

In order to have successful projection of image over all part of the surface, the surface must be star shape with respect to the projector. A surface has a star shape if any straight line from projector toward the surface intersects the surface only maximum of once [19]. This characteristic of the surface introduces limitations on the distance between the projector and the surface, slope of surface, and elevation of surface. The platform must support 10 cm elevation difference between the poles. From the previous section, we know that the number of poles beneath the platform is 25, and the size of the platform is 60cm x 60cm. Hence, assuming the projector is located in the middle of the platform at the top, the farthest place from the projector is in the direction of the surface diagonal. In this case, the distance between two neighbor poles in this direction is $10\sqrt{2}$ cm. The figure below presents the side view of this situation.

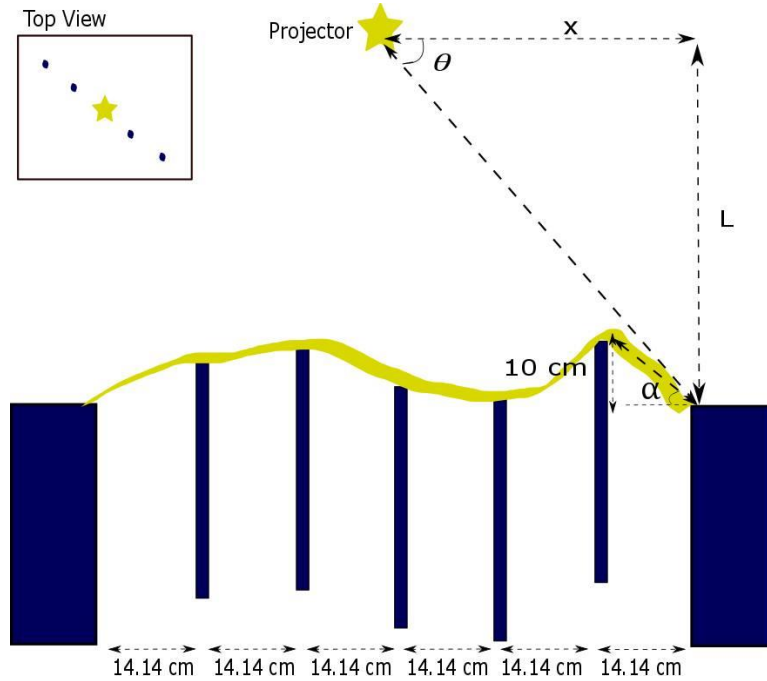


Figure 8. Star Shape Surface Analysis

The following analysis shows in detail how the specifications can be met with specifying of distance of the projector from the surface.

$$\begin{aligned}\theta &> \alpha \\ \tan(\theta) &> \tan(\alpha)\end{aligned}$$

$$\frac{L}{x} > \frac{\text{Elevation difference between two neighbour poles}}{\text{distance between poles}}$$

$$L > 30 \text{ cm}$$

Hence, as long as the distance of the projector from the surface is greater than 30 cm, the elevation specification and also the star shape surface limitation are met.

3.3.3.2 Projected Image Resolution

The image processing techniques used for the projects does not change the number of pixels. In fact, either the intensity or the position of the pixels with respect to the raw image is modified. Hence, the number of pixels (resolution) of the processed image does not vary with respect to the input resolution. Hence, as long as the resolution of the input image meets the specifications, the output image meets the resolution specification.

3.3.4 Image Processing Techniques and Algorithms

3.3.4.1 Image Access

The image can be accessed easily through MATLAB after file is selected by the user over the user interface. The following is the code for this matter.

```
IMAGE = imread('path of the image stored'); % read and store RGB pixel information of input image
```

3.3.4.2 Curvature of Surface

The image processing techniques used in this project needs the curvature of the surface. The curvature of the surface can be calculated by finding the first and second fundamental form. First fundamental form is the inner product on the tangent space of the surface [20] which is as follows:

$$E dx^2 + 2F dx dy + G dy^2$$

The second fundamental form is a quadratic form on the tangent plane in the three dimensional Euclidean space [21] which has the following form:

$$L dx^2 + 2M dx dy + N dy^2.$$

Hence, the mean curvature can be found from the following formula [22]

$$K_m = \frac{1}{2} * \frac{L * G + N * E - 2 * M * F}{E * G - F^2}$$

The following code shows how the curvature of the surface is calculated from elevation data.

```
[X,Y,Z] = read and store input elevation data here from User Interface;  
[K,H] = surfature(X,Y,Z); % calculate and store the Gaussian and Mean curvature, respectively.
```

The surfature function shown below is from source [23].

```
function [K,H,Pmax,Pmin] = surfature(X,Y,Z),  
% SURFATURE - COMPUTE GAUSSIAN AND MEAN CURVATURES OF A SURFACE  
% [K,H] = SURFATURE(X,Y,Z), WHERE X,Y,Z ARE 2D ARRAYS OF POINTS ON THE  
% SURFACE. K AND H ARE THE GAUSSIAN AND MEAN CURVATURES, RESPECTIVELY.  
% SURFATURE RETURNS 2 ADDITIONAL ARGUEMENTS,  
% [K,H,Pmax,Pmin] = SURFATURE(...), WHERE Pmax AND Pmin ARE THE MINIMUM  
% AND MAXIMUM CURVATURES AT EACH POINT, RESPECTIVELY.  
% First Derivatives  
[Xu,Xv] = gradient(X); [Yu,Yv] = gradient(Y); [Zu,Zv] = gradient(Z);  
% Second Derivatives  
[Xuu,Xuv] = gradient(Xu); [Yuu,Yuv] = gradient(Yu);  
[Zuu,Zuv] = gradient(Zu); [Xuv,Xvv] = gradient(Xv);  
[Yuv,Yvv] = gradient(Yv); [Zuv,Zvv] = gradient(Zv);
```

```

% Reshape 2D Arrays into Vectors
Xu = Xu(:); Yu = Yu(:); Zu = Zu(:);
Xv = Xv(:); Yv = Yv(:); Zv = Zv(:);
Xuu = Xuu(:); Yuu = Yuu(:); Zuu = Zuu(:);
Xuv = Xuv(:); Yuv = Yuv(:); Zuv = Zuv(:);
Xvv = Xvv(:); Yvv = Yvv(:); Zvv = Zvv(:);
Xu = [Xu Yu Zu]; Xv = [Xv Yv Zv];
Xuu = [Xuu Yuu Zuu]; Xuv = [Xuv Yuv Zuv]; Xvv = [Xvv Yvv Zvv];
% First fundamental Coefficients of the surface (E,F,G)
E = dot(Xu,Xu,2); F = dot(Xu,Xv,2); G = dot(Xv,Xv,2);
m = cross(Xu,Xv,2); p = sqrt(dot(m,m,2)); n = m./[p p p];
% Second fundamental Coefficients of the surface (L,M,N)
L = dot(Xuu,n,2); M = dot(Xuv,n,2); N = dot(Xvv,n,2);
[s,t] = size(Z);
K = (L.*N - M.^2)./(E.*G - F.^2); K = reshape(K,s,t); % Gaussian Curvature
H = (E.*N + G.*L - 2.*F.*M)./(2.*(E.*G - F.^2)); H = reshape(H,s,t); % Mean Curvature
Pmax = H + sqrt(H.^2 - K); Pmin = H - sqrt(H.^2 - K); % Principal Curvatures

```

The following figure is an example of a curved surface and the corresponding curvature (please note that the elevation data and coordinates used for this example do not necessarily meet the specifications of the design).

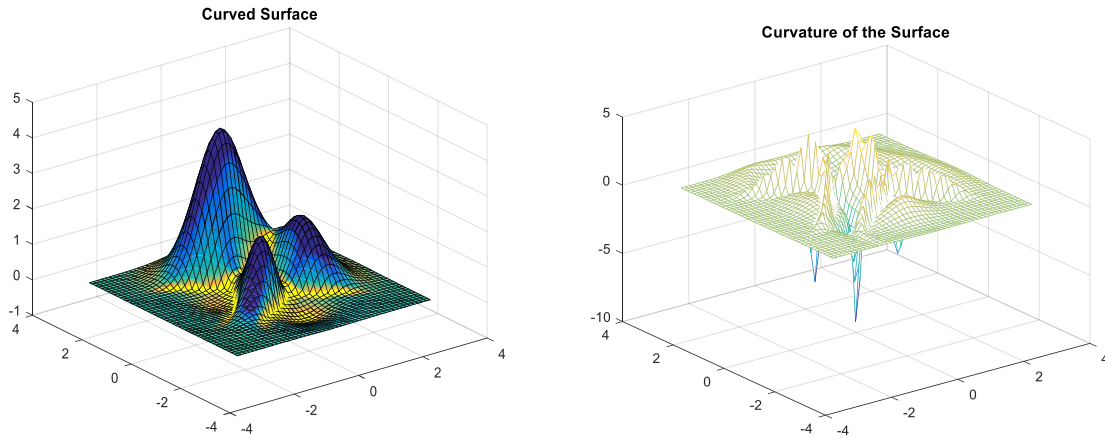


Figure 9. Example of Curved Surface and its Curvature.

3.3.4.3 Surface Normal

The surface normal is also essential in the image processing algorithms in this project. The surface normal is used to calculate the angle between the direction of the projected image and the surface normal. This angle is later used in Lambertian strategy for elimination of distortion. There are two methods to calculate the surface normal. Both methods give the similar results with some slight differences near the edges of the surface due to the difference in assumption of boundary conditions. First method uses the built-in MATLAB function of 'surfnorm'. The second method is

derived from the concepts of geometry. In fact, defining the surface as a function $f(x, y)$ where x and y are independent variables, the following implicit form can be derived [24].

$$F(x, y, z) = z - f(x, y) = 0$$

The normal to the surface is the gradient of this implicit form. Hence, the normal can be calculated from the following formula where i , j and k are the x , y and z unit vectors [24].

$$\nabla F(x, y, z) = k - \frac{\partial f(x, y)}{\partial x} i - \frac{\partial f(x, y)}{\partial y} j$$

The methods coded in MATLAB are shown below.

```
% Method 1
[Nx,Ny,Nz] = surfnorm(X,Y,Z);
% Method 2
[fx,fy]=gradient(z,dt,dt); % dt is the spacing between coordinates
a=sqrt(fx.^2+fy.^2+1);
Nx=-fx./a;
Ny=-fy./a;
Nz=ones(N)./a; % N is the size of the surface points
```

3.3.4.4 Angle between Surface Normal and Projection Line

For the calculation of the angle between the direction of projection and the surface normal, two methods are developed. The first method assumes that the projector is far enough ($L > 4$ m) so that the projected lines are all almost parallel to one another and normal to the flat surface. This this method the angle is calculated as follows:

```
angle = acosd(Nz);
```

In the second method, it is assumed that the projector is close to the surface, so the approximation of the first method is not valid anymore. In this case, the unit vector from the surface in the direction of projector is derived, and then the angle is calculated. The steps for this analysis are shown below.

```
% position of the projector (d is the distance of the projector from the surface)
projectorPos = [0;0;d] ;
angle = zeros(size(Z,1),size(Z,2));
for i = 1:size(Z,1)
    for j = 1:size(Z,2)
        u = projectorPos - [X(i,j);Y(i,j);Z(i,j)];
        v = [Nx(i,j);Ny(i,j);Nz(i,j)];
        angle(i,j) = acosd(dot(u/norm(u),v));
    end
end
```

The following figure shows the angle in degrees between the direction of projection and surface normal from both methods, assuming the distance between the projector and surface is 30cm.

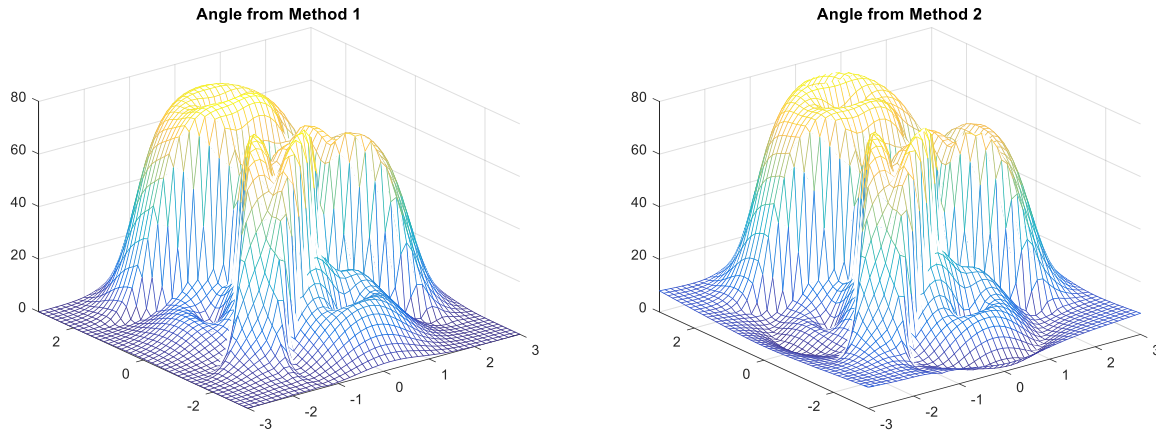


Figure 10. Angle between the projection line and surface normal.

3.3.4.5 Lambertian Model

As known, the projection of raw image into the curved surface results in distortion of the image. There are many different ways to tackle this distortion. Some of the strategies involve complicated differential geometry. In our project, we choose a simple strategy (Lambertian model) which is proven to be effective to remove some part of the distortion. The reflection of the image from the surface is a type of diffuse reflection. The apparent brightness of diffusely reflecting surfaces can be ideally estimated by Lambert's cosine law [25]. In fact, the luminous intensity observed from the surface is proportional to the cosine of the angle between the normal of the surface and the direction of projected image [26]. For the projection of image, this non-uniformly distribution of intensity is taken into consideration for the following reasons. The projector projects the image in a concave shape. As the projector is mounted closely on top of the surface, this concave projection results in decrease of luminous intensity toward the edges of the surface. In addition, the elevation and slope of the surface affect the angle between the projected line of image and the normal to the surface. To compensate this loss of intensity, each individual pixel of the image is divided by the value of this cosine angle so that the pixels which appear to be less luminous are projected with higher intensity. In calculation of this angle, method 2 of calculating the angle proposed at previous sub section gives a more accurate result. The following code shows the steps taken for this matter.

```
IMAGE = IMAGE ./ cosd(angle);
IMAGE = (255/max(IMAGE)) * IMAGE; % this is to fit the range of RGB image to uint8.
```

To justify that this method is effective in compensate the loss of intensity introduced by the slope of the surface with respect to the projector, the following test is performed. The projector is positioned at distance of 1 meter in 30 degrees angle with respect to a flat surface normal. A solid color image with uniform intensity is projected to the surface. The following code is used to apply the Lambertian model to the image before projection.

```
d = 1; % 1 m is normal distance of surface from projector
pangle = 30 % the angle of the projector with respect to the surface normal
```

```

surface_width = 0.6; % 0.6 m
x1 = tand(pangle) * d;
x2 = x1 + surface_width;
IMAGE = imread(pth); % read the image to 3D matrix X x Y x RGB
N = size(IMAGE); % take the size of the image
x = linspace(x2, x1, N(2));
theta = atan(x/d);
m = 1./cos(theta); % create 1D mask from lambertian model
M = repmat(m,[N(1) 1 3]); % add N(1)rows, 1 duplicate (none), 3 depth layers
In = M.*double(IMAGE); % manipulate the intensity of the image by M mask values
In = uint8((255/max(In(:)))*In); % divide by max so rgb values are in range of 0 to 255

```

The following figure presents the projected raw image on the left, and the projected processed image on the right. It can be seen on the left image that the image loses its intensity as it goes from right to left. Furthermore, it can be seen on the right image that the intensity is almost uniform all over the image.

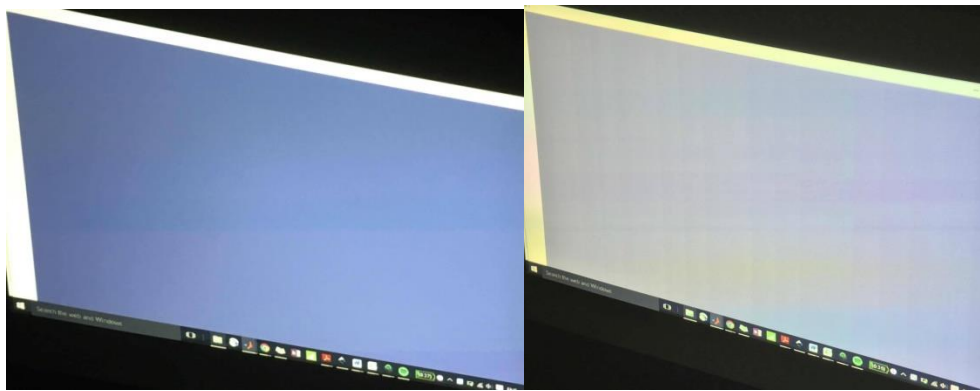


Figure 11. Lambertian Model Effect on Image.

3.3.4.6 Deformation

The projection of raw image into the surface results in deformed image due to the curvature of the surface. Image warping is the transformation of image such that points are mapped to some other points while keeping their original colors [27]. In this project, this type of transformation is used to eliminate the distortion of the image as if it is projected onto a curved surface. The idea is that the curvature of the surface is used to build the transformation mapping of the raw image into an image which has the complement curvature of the surface. Hence, projection of this image into the curved surface results in a projected image which appears flat to the viewer looking into the platform from the top.

3.3.4.7 Projection

The projection of the image onto the surface is done through MATLAB built-in function of 'imshow'. The following code is for displaying of the image as full-screen.


```
imshow(IMAGE)
hFig = gcf; hAx = gca;
set(hFig,'units','normalized','outerposition',[0 0 1 1]); % set the figure to full screen
set(hAx,'Unit','normalized','Position',[0 0 1 1]); % set the axes to full screen
set(hFig,'menubar','none') % hide the toolbar
```

3.4 Desktop Application

One of the components of desktop application is the front-end interface that users first see, before the process of landscape simulation begins. Using the application, the users verify and transmit the landscape information to be simulated as a CSV file. Another alternative file type considered is a CAD file or a converted CAD file (elevation information extracted into .csv), but it is neglected in this report because it is not an essential specification.

When the input file is selected, the content of the file is verified by the Input Limitation and Conversion Algorithm. Once validated, the algorithm converts the elevation data into the “on” time of the stepper motor correspondingly, which ultimately controls the height of the flexible surface. Since the algorithm does not require external data (database), front-end development is required only. Therefore, choosing a capable front-end tool(s) is critical. The major concerns of the front-end interface (or UI) design are:

- Learning must be minimized; implement using familiar tools
- Many examples must be available to minimize development time
- UI must be connected to the Input Limitation and Conversion Algorithm
- Tools must be open source; associated cost is zero
- A CSV can be uploaded

3.4.1 UI: First Iteration

Qt is a cross-platform application framework, utilizing standard C++, used for developing software applications that can run on various software and hardware platform, while still being native with the capabilities and speed [28]. In addition, Qt Open Source, identical to the commercial version, is available to students and individuals, as long as the terms of the GPL or LGPL are complied. Therefore, the development is first attempted with Qt. Installation of Qt includes the IDE specific to Qt, Qt Creator, which allows users to create GUI (front-end page) by dragging and dropping pre-defined objects including buttons and labels, leading to a quick development of the front-end page, as shown in Figure 12.

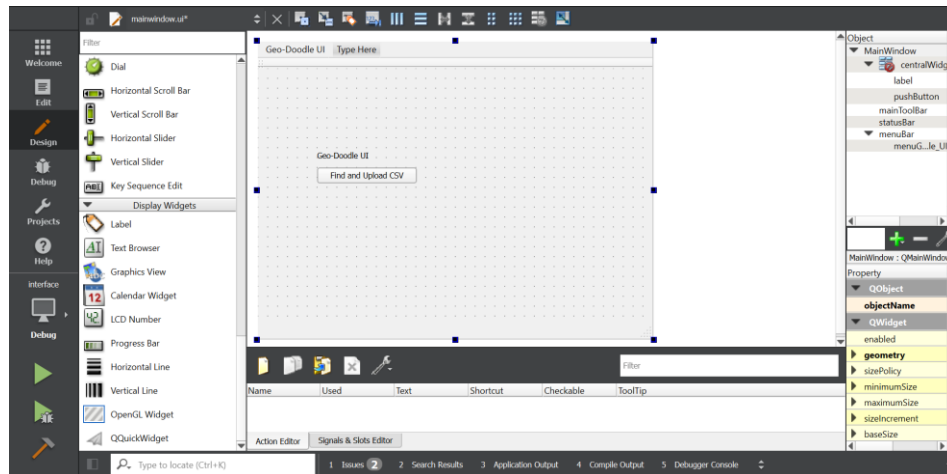


Figure 12. UI design of QT

Upon completion of the front-end display, a code is written in C++ to activate the uploading of a CSV file (or a text file) when the button created is clicked. The code is shown in Figure ##.

```
#include "mainwindow.h"
#include "ui_mainwindow.h"
#include<QFileDialog>

MainWindow::MainWindow(QWidget *parent) :
    QMainWindow(parent),
    ui(new Ui::MainWindow)
{
    ui->setupUi(this);
}

MainWindow::~MainWindow()
{
    delete ui;
}

void MainWindow::on_pushButton_clicked()
{
    QString filename=QFileDialog::getOpenFileName(
        this,
        tr("Open File"),
        "C:/"
        "CSV (*.csv); Text File (*.txt)"
    );

    QMessageBox::information(this,tr("File Name"), filename);
}
```

Figure 13. C# Code of Getting csv File as Input

The code shown in Figure 13 simply imports a library called QFileDialog, one of the Qt library components, and uses one of its functions to allow users to select a desired file from the file explorer. Then, returns a message box that displays the name of the chosen file. The code compiles without the message box, and does not compile with the code fragment corresponding to the message box. The error is due to the function called, QMessageBox,

not being recognized. When the problem occurred, the team quickly realized that none of the members are familiar with the tool and also that there are limited number of examples and sources available in public. Since minimal development time and high efficiency are very important, another solution is considered.

3.4.2 UI: Second Iteration

The alternative to Qt is a web-based tool, HTML. Developing a web application using HTML does not require an installation of any software (other than a web browser to visualize the output); zero cost associated with local development. Besides, a limited number of tags involved in HTML leads to minimized learning time required for the team members who are not familiar with the tool. The front-end page developed using HTML is presented in Figure 14.

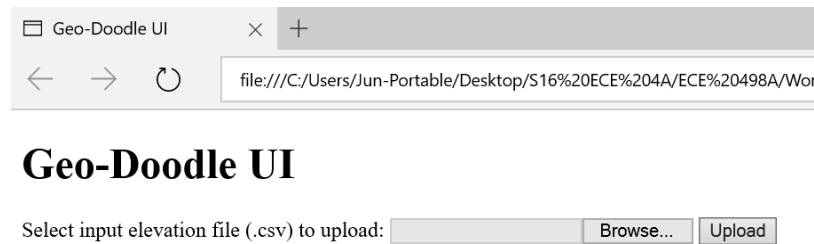


Figure 14. UI design of Web

The front page also utilized an example of PHP script found in public [29] to allow prompting a file explorer when “Browse...” button is clicked. Along with the PHP script example, we found that there are many more examples and sources available for HTML than there are for Qt. For example, Google search indicates that there are about 6 billion results for html and about 98 million results for qt. In addition, a web application is cross-compatible with various operating systems, web browsers, and script languages, which allows various options (JavaScript is preferred because all of the team members have an experience) for connecting with and development of the algorithm. Thus, it is one of the best tools for the team, composed of members with different technical background.

3.4.3 Input Limitation and Conversion Algorithm

The algorithm checks that the input CSV file is a 5x5 matrix: 5 rows of data stream, consisted of 5 elements (5 columns), which corresponds to the number of stepper motors and their allocation across the platform surface. If the input file is not in the required format, the algorithm indicates the incompatibility to the UI. The pseudo-code of the algorithm is shown in the following; running time is $\Theta(n)$:

```
row=1; column=0;
while(read()!=null){
    column++; nextElement=gotoNextElement_and_Read();
```

```

if(nextElement==null && column!=5){
return text("Column not equal to 5: "+column+" at Row:"+row ;}
elseif(nextElement==null && column==5){
    nextLine_FirstElement=gotoNextLine_FirstElement_and_Read();
    if(nextLine_FirstElement!=null && row<=4){row=row+1; column=0;}
    elseif(nextLine_FirstElement==null && row==5){return text("Success");}
    elseif(nextLine_FirstElement!=null && row==5){return text("Bad: More than 5 rows");}
}
}

```

The alternative to a 5x5 input arrangement is a linear arrangement of 25 elements. However, the linear arrangement does not distinguish other possible combinations that compose 25 elements (for example, 4 elements for the first row, 6 elements for the second row, and 5 elements for the other 3 rows). Thus, it is not considered.

Another important responsibility of the algorithm is the conversion of elevation data into the “on” time of each of the stepper motors, which ultimately controls the elevation of surface. Since the maximum height achieved by the surface is 10 cm, the input elevation elements can be normalized using the following equation:

$$\text{Normalized Element Value} = \frac{\text{Value of Each Element}}{\text{Value of Highest Element}} \times 10 \quad (1)$$

Then, the normalized elements are converted into the motor running time using the following equation:

$$\text{Time} = \frac{\text{Normalized Element Value}}{10} \times \text{Time Required to Lift Surface by 10 cm} \quad (2)$$

Once the conversion is completed, the 5x5 matrix of computed motor running times is transferred to the microcontroller through USB, to let the action begin.

3.4.4 Actuator Control

The actuator chosen in Section 3.2 is a 2-phase bipolar stepper motor, and driving such motor is essentially a circular iteration of attraction between coils excited by change in current flown through (induced by ΔV between the coil ends) and a bipolar magnet. A 2-phase bipolar stepper motor contains two coils that either of their ends is grounded, thereby requiring 4 inputs which the states are correlated to each other. The rotation steps with respect to the input states are illustrated in Figure 15 [30].

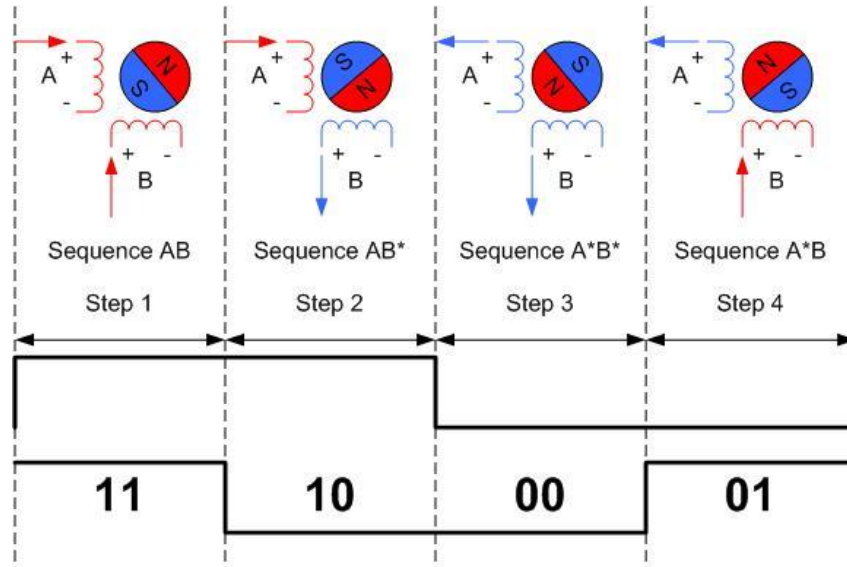


Figure 15. Rotation of 2-phase Bipolar Stepper Motor with respect to inputs.

A more practical implementation of the input waveforms is allocating some dead time in between the rotation steps (or transitions), as shown in Figure 17 [31].

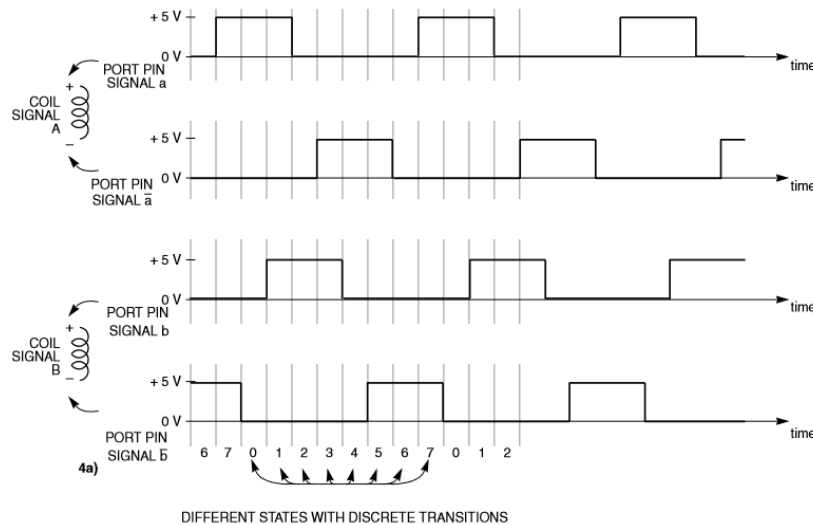


Figure 16. A practical version of input waveforms demonstrated in Figure 5.

In all practical situations, signals associate rise and fall times, which cause interference with the adjacent input states if such times are larger than the dead time allocated. If the interference is not mediated, it causes a propagation delay for each signal line. Consequently, the voltage difference across an inductor reduces (worst-case scenario, it may reduce to 0 V at some point in time, where it should be the maximum swing voltage), which reduces the change in current. Ultimately, the attraction force decreases, thereby increasing the chance of missing a stepping action. The time is investigated once the stepper motors are retrieved and become available for experiments.

The hardware acquired are two of the Arduino Mega boards, borrowed from the ECE Inventory. Each board has 54 digital I/Os, thus one of the boards controls the 13 motors and the other controls the 12 motors, to control total of 25 motors. Since each I/O pin is capable of supplying 20 mA, each motor that requires current of 500 mA at (4 ~ 6) VDC for operation is driven by an external driver that is capable of providing 600 mA [32].

4. Discussion and Project Timeline

4.1 Evaluation and Final Design

The selection of surface meets the specifications in terms of color and elasticity. The overall weight of the system is around 8 kg (5 kg platform, 2.7 kg projector) which is lower than the maximum weight (20 kg) specified. Moreover, a minimum projector distance from surface is specified in the design such that projection of image would cover the whole surface area. The image processing algorithms are designed to meet the specification of distortion elimination. The design of the user interface meets the specifications in terms of the monitoring the input to make sure the system gets a valid input. The elevation actuators integrated on the platform is 10cm which meets the maximum surface height specification. The total cost of the components is estimated to be \$317 (\$300 Actuators + \$17 surface) which meet the budget in specifications.

4.2 Use of Advanced Knowledge

Our project combines knowledge from ECE 224, which covers microprocessors and embedded systems, and ECE 455, which covers embedded software development to implement actuator controlling system, which fetches elevation data from .csv files and converts numerical values to appropriate voltage level. Knowledge of image processing is from ECE 417, which is taught by Professor Michailovich, who is also the consultant for this group. Image processing is a key component of this project, as it enables accurate representation of the geographical model by tweaking the original image.

4.3 Creativity, Novelty, Elegance

Traditional method of representing a geographic landscape is either to build one manually using materials such as clay, or to create a 3D model using computer software. While both methods have their own pros, they also have their cons. Former method provides a static solution, which cannot be altered once finished. Latter method offers freedom to build any model of their interest, but presenting the model may not be much engaging. Lots of companies today work on VR (virtual reality) technology, and VR is a possible solution to this design goal. However, virtual technology has a limitation since it is only perceived through vision and requires users to wear VR devices. This project enables dynamic representation of geographical landscapes via a set of actuators, which frees audiences from requiring any external devices and also allows them to touch the generated surface.

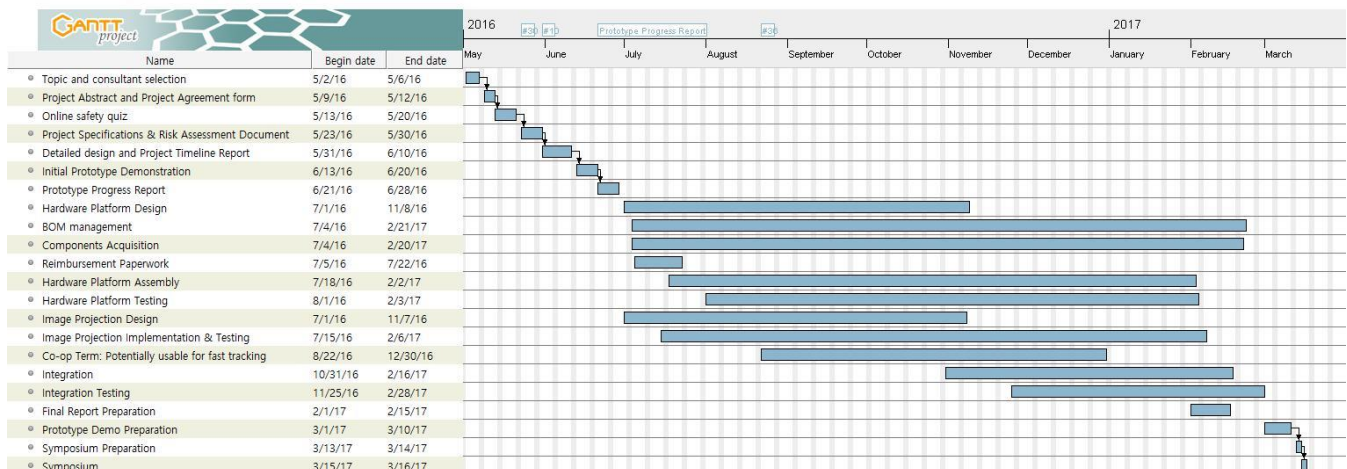
4.4 Student Hours

Student	Hours
	70
	72
	54
	67
	70

4.5 Potential Safety Hazards

This design uses standard North American residential electricity of 110 V and 4 A. Electric current of 100 mA is enough to cause human muscle contraction, and 200 mA of current is potentially fatal. Therefore, if design is faulty or not properly insulated, it may electrically shock users or damage itself. It also must be noted that this project involves numerous actuators. Although they may be small in size, they may possess enough power to catch and damage body parts.

4.6 Project Timeline



Majority of tasks are to be done during the spring term of 2016 and the winter term of 2017. The deadlines are either determined by school or the team in the most reasonable manner to successfully be prepared for upcoming events. The tasks given by school such as product specification and detailed design documentation are chained as a single critical path since they are to be completed one by one. However, majority of the critical tasks are to be in progress simultaneously in order to complete the project on time. Therefore, plenty of time is assigned to design, testing, and integration phase of this project and there is no critical path in entire prototyping phase of this project. Additionally, majority of team members are staying in Ontario during their coop term, so it may be possible to arrange meetings during the term to make progress in ideal case. Finally, Demo and symposium preparation start when integration is finished in order to successfully deliver the prototype to public.

5. References

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